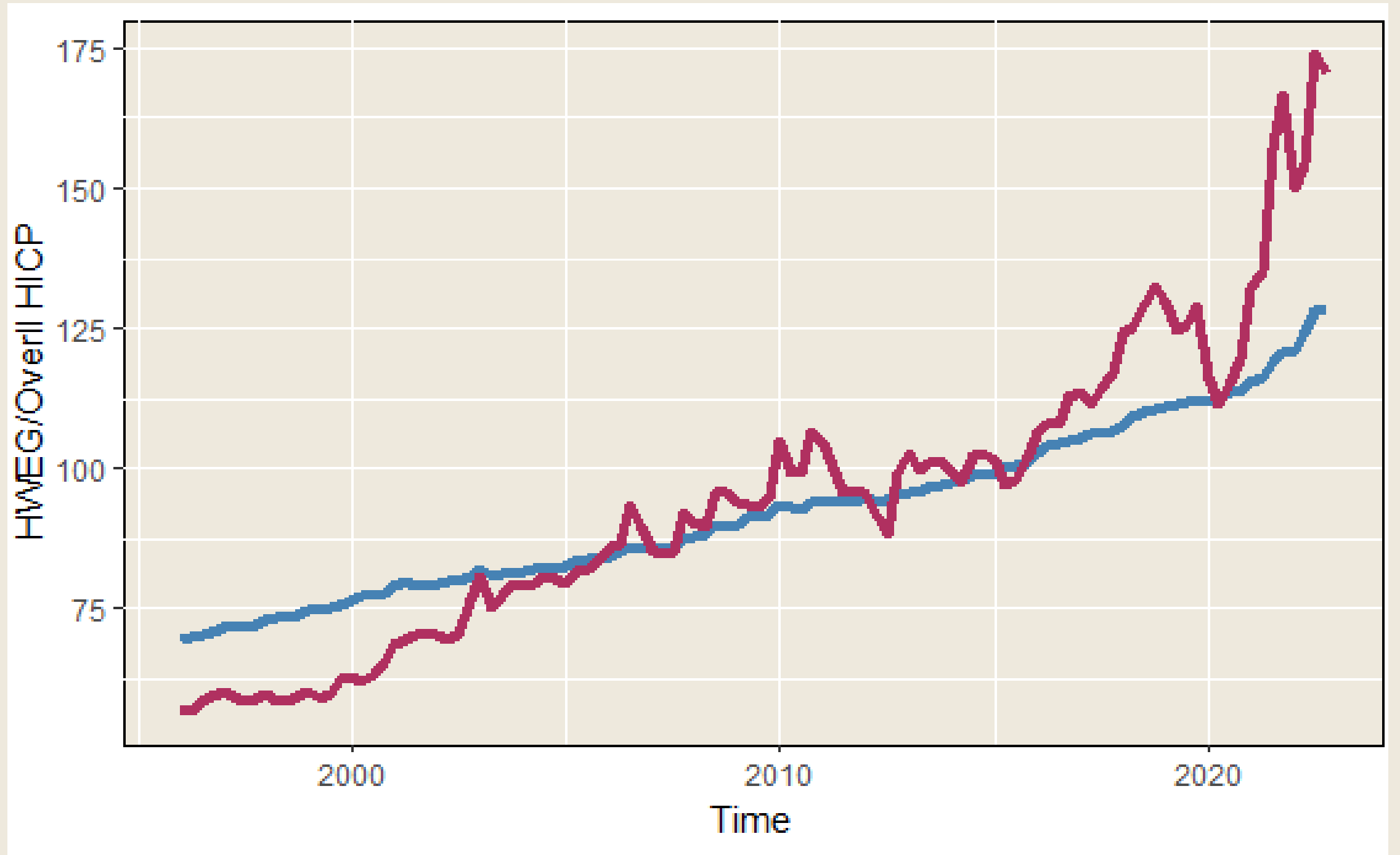
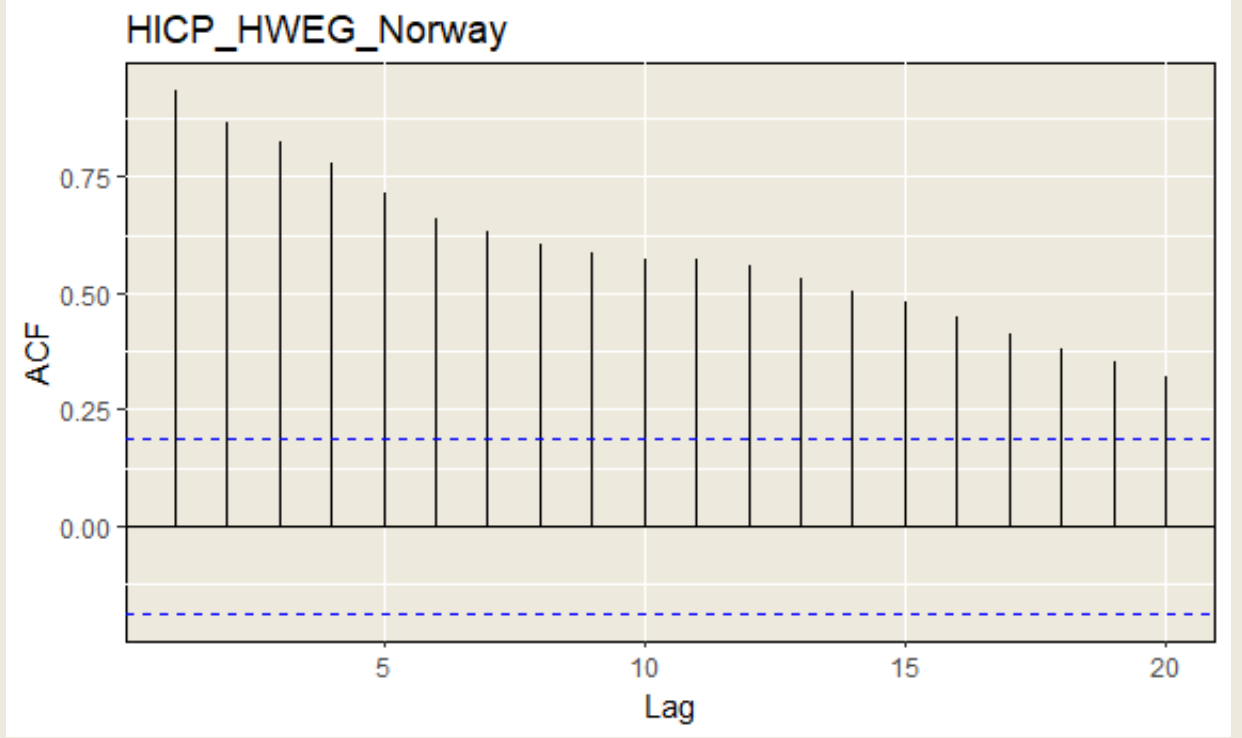
Modelling Norway's Cost Of Living Crisis through HICP Data

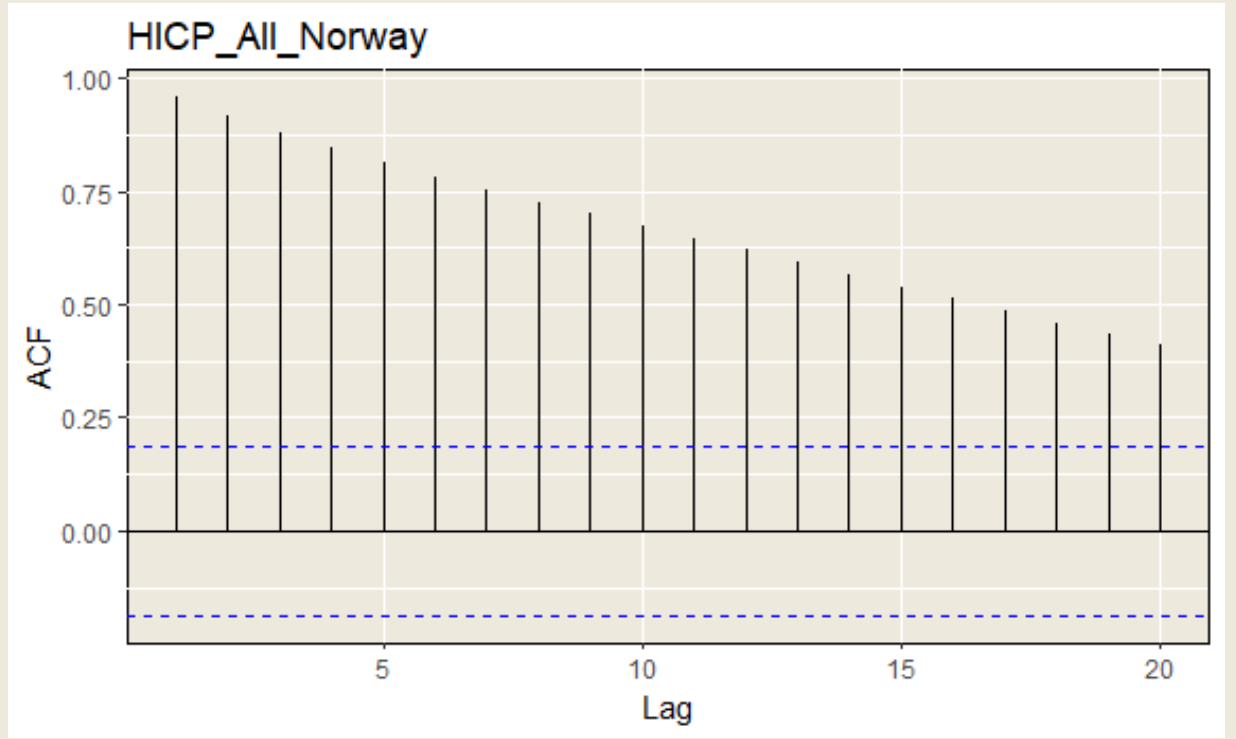
01. Introduction and Objective

It's been a cold winter in Europe. Soaring energy prices and wage-price spirals have meant inflationary pressure, while repeated rate hikes and monetary policy tightening impose recession fears. Rising prices can have real effects on livelihoods. Accurate inflation forecasts are vital in influencing policy decisions. In this treatise, we explore the cost of living crisis in Norway and converge on the best model for forecasting what lies ahead for the land of fjords, Vikings, and Haaland!

We start by plotting quarterly indices harmonized index of consumer prices (HICP) for overall(blue) and for housing, water, electricity, and gas(HWEG)(maroon) against time.







Both indices seem to display an increasing trend over time. The HWEG HCIP index seems to display more volatility in recent years and has risen more sharply than the overall HCIP index. The sharp dip around 2020 could be caused by reduced demand for fuels during the lockdown, while the sharp upturn could reflect the rising prices of energy following the supply-side shock caused by the Ukraine invasion. Rising food and energy prices could account for the continued rise.

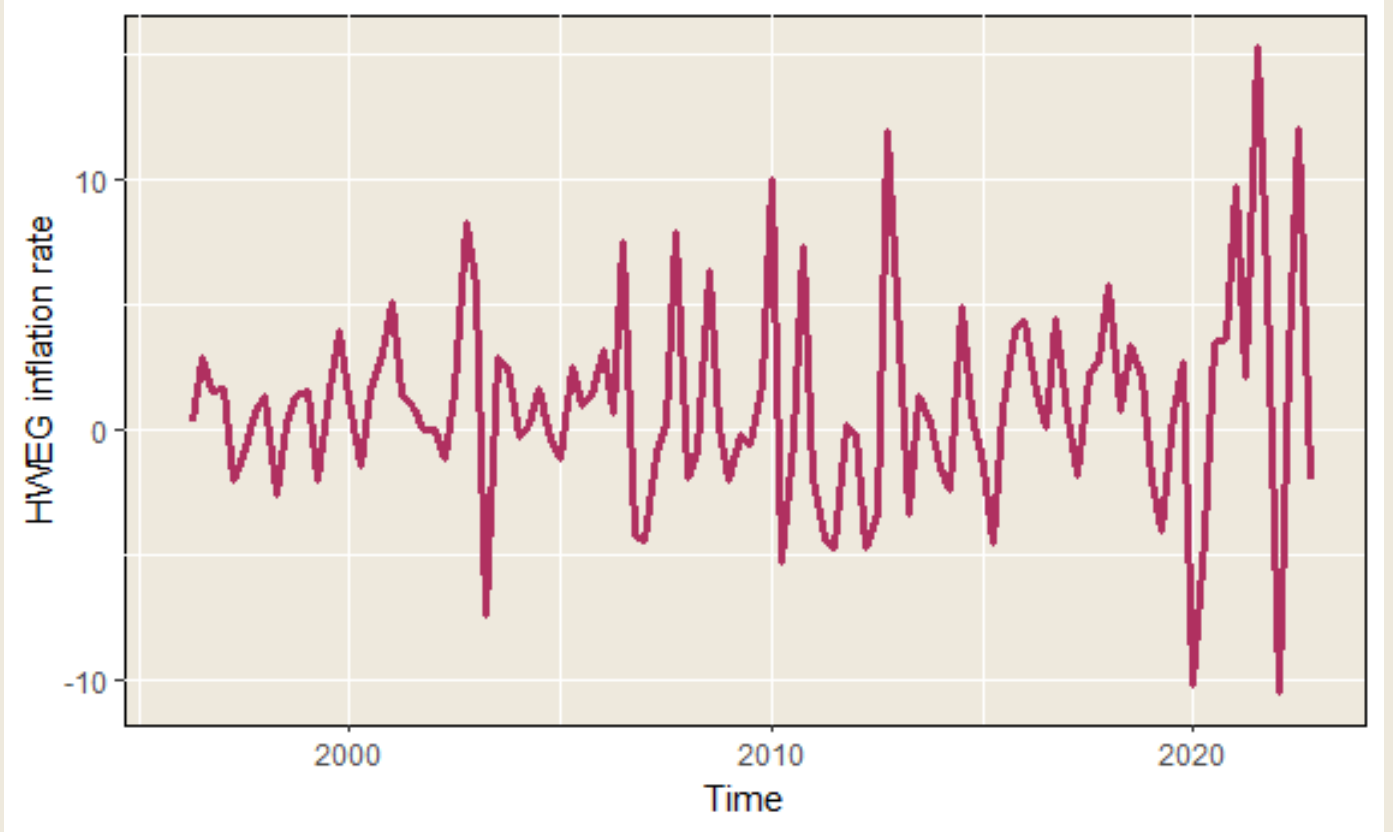
2. Modelling and Forecasting

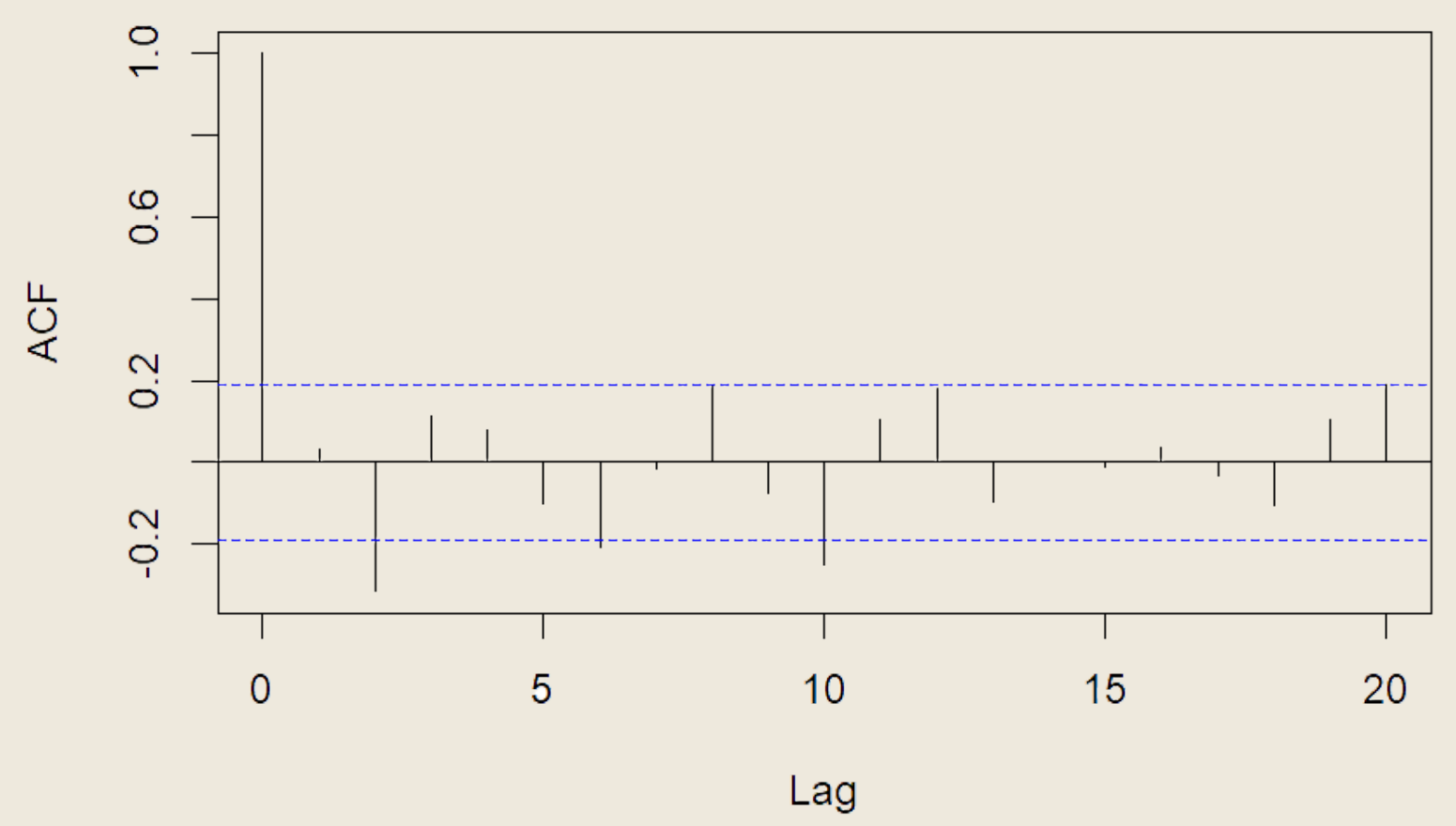
To be able to apply statistical theory and forecast with time series, the time series must be ***stationary***. A time series Yt with var(Yt) <infinity, is weakly stationary when: **1.** E(Yt) is constant **2.** Var(Yt) is constant **3.** *cov Yt , Yt+j* only depends on *j* and not *t* ie it is time independent.

Here the series has no constant mean(it trends upward) and the ACF plot drops off slowly implying that the autocorrelation is time-dependent. We must stationarise the data.

2.1. HWEG HICP Data

We start with the HWEG data and stationarise the series by taking the log of the first difference. The data appears to be stationary as the ACF drops off quickly to 0 and the Dickey-fuller test returns a low p-value implying we can reject the null hypothesis of non-stationarity at the 5% significance level. (Fig below). We can begin to build models





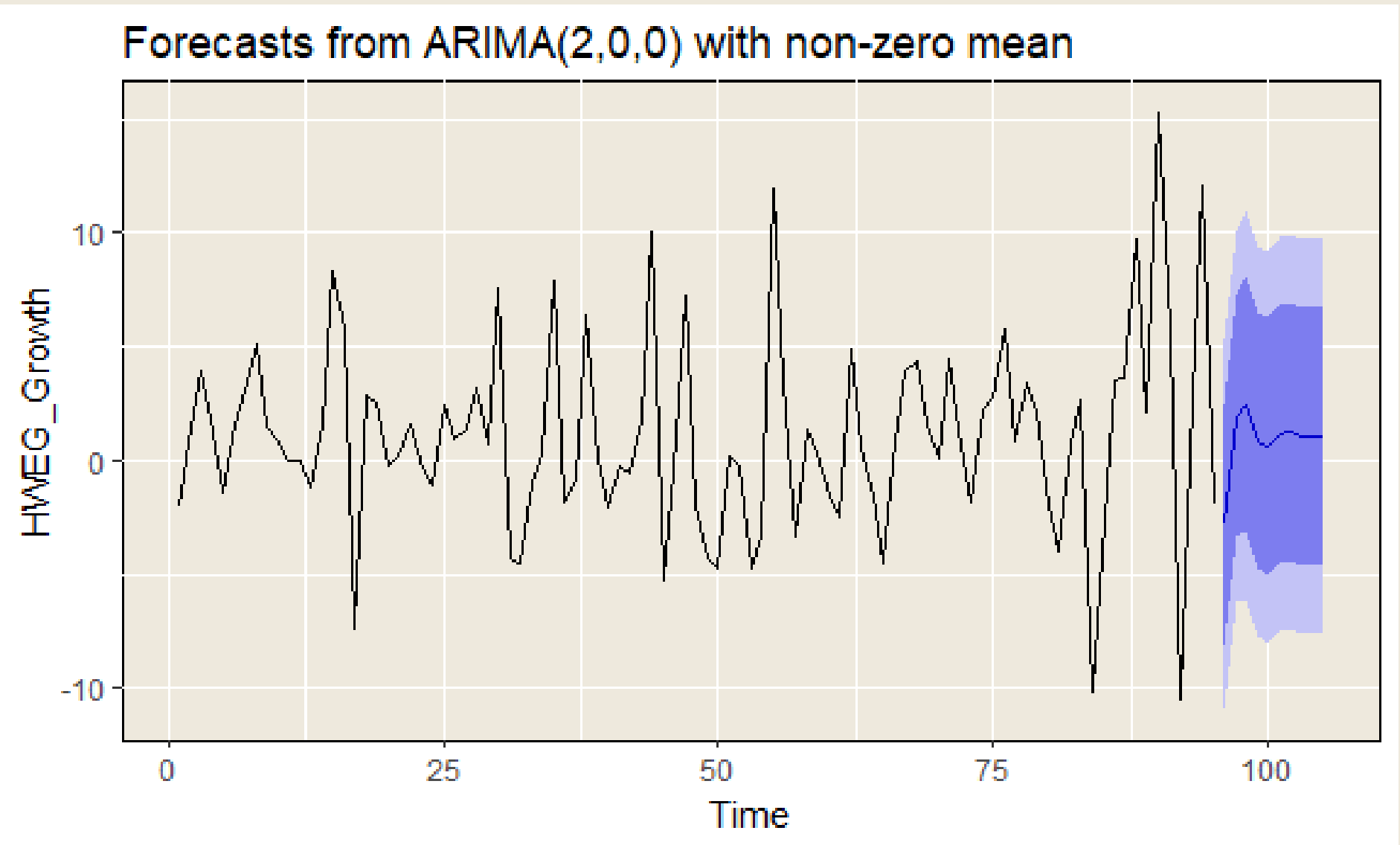
2.1.1. Methodology and Findings

We use autoregressive models that express the dependent variable of a time series Yt as a function of its lags. We'll build AR(1) to AR(12) and use the Bayesian Information Criterian (BIC) to choose the best model. We will then test for the presence of outliers using the 1.5IQR rule and see how our regression output and forecasts change when we drop outliers. Lastly, we will use ARMA which aims to preserve the predictive power of the AR model while improving on the plausibility of the assumption behind the error term ut.

Stationary first order regressive model(AR1) : Yt = B0 + B1Yt-1 + ut where ut is a white noise error process with E(ut)=0, var(ut)=s^2 and cov(ut, ut-j)=0

Findings of models with the lowest BIC in each category are summarised below:

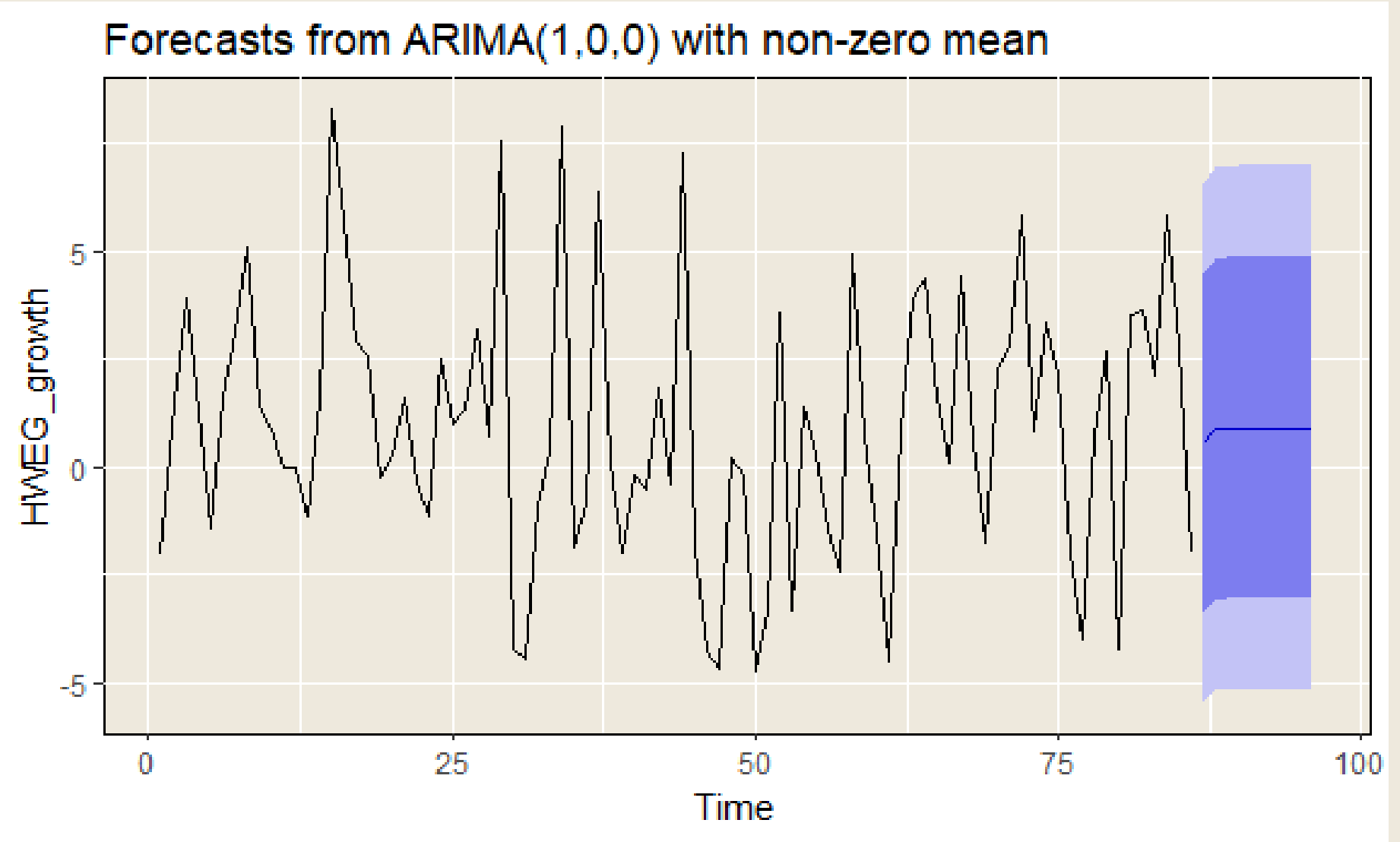
WITH OUTLIERS



AR2

- **BIC = 554.96**
- **Coeff : ar1: 0.0482, ar2: -0.34 (|*B*|<1)**
- **RMSE : 4.074**
- **Point forecast using estimated coefficients: Yt+1|t = *-2.8317***
- **95% CI : (-10.9464, 5.2829)**

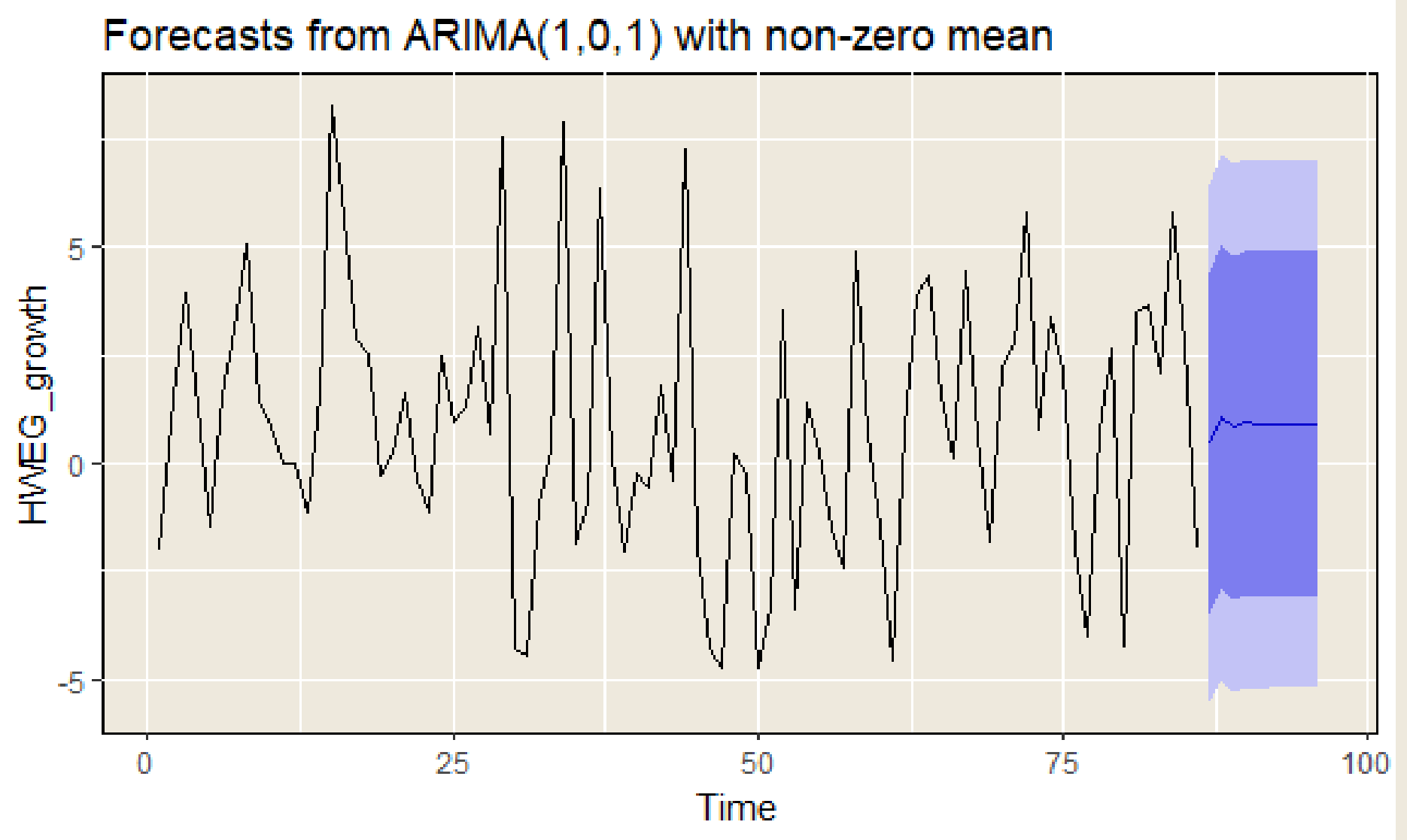
WITHOUT OUTLIERS



AR1

- **BIC = 448.4819**
- **Coeff : ar1: 0.129(|*B*|<1)**
- **RMSE : 3.03656**
- **Point forecast using estimated coefficients: Yt+1|t = *0.532374***
- **95% CI : (-5.489601, 6.554349)**

ARMA WITHOUT OUTLIERS



ARMA(11)

- **BIC = 451.24**
- **Coeff : ar1: -0.4320, ma1: 0.6065**
- **RMSE : 3.006**
- **Point forecast using estimated coefficients: Yt+1|t = *0.4568701***
- **95% CI : (-5.540312, 6.454053)**

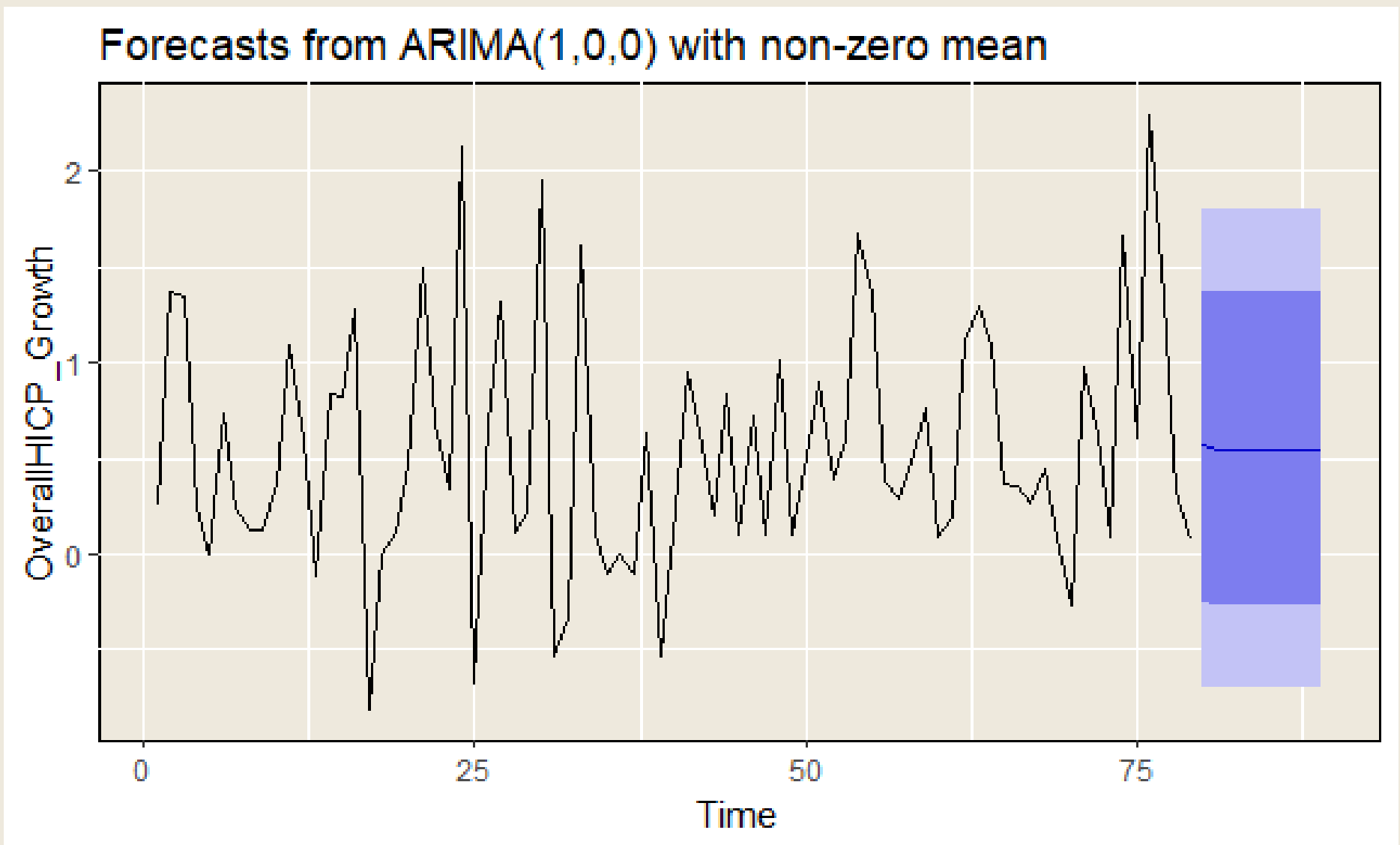
We find that the BIC values for models built without outliers are lower. *Sometimes the simple model is best*, in this case, the AR1 model built on data without outliers seems to have the lowest BIC. The AR1 forecasts that the HWEG_HICP is *expected to increase by 0.532%* in Yt+1. The expected Index value in period T+1(Yt+1) is **179.968** using Yt+1 **= e**xp(0.532/100 + log(lag(HWEG_HICP)). The 95% confidence level for the expected index value is **(179.4205, 180.5495)** **|B| < 1** as we would expect in the case of stationarity. It implies a 1% growth in the HICP_HWEG index in period t-1 will cause a 0.532% growth in HICP_HWEG in period t.

2. 2 Overall HICP Data

We repeat the above procedure for overall inflation HICP. However, we find that even on taking the log of the difference, the data isn't stationary and we can't reject the null hypothesis of the Dickey-Fuller Test at the 5% significance level (p-value = 0.13). However, removing the outliers seems to solve the problem. So here we will only look at AR and ARMA models without outliers. Results shown below:

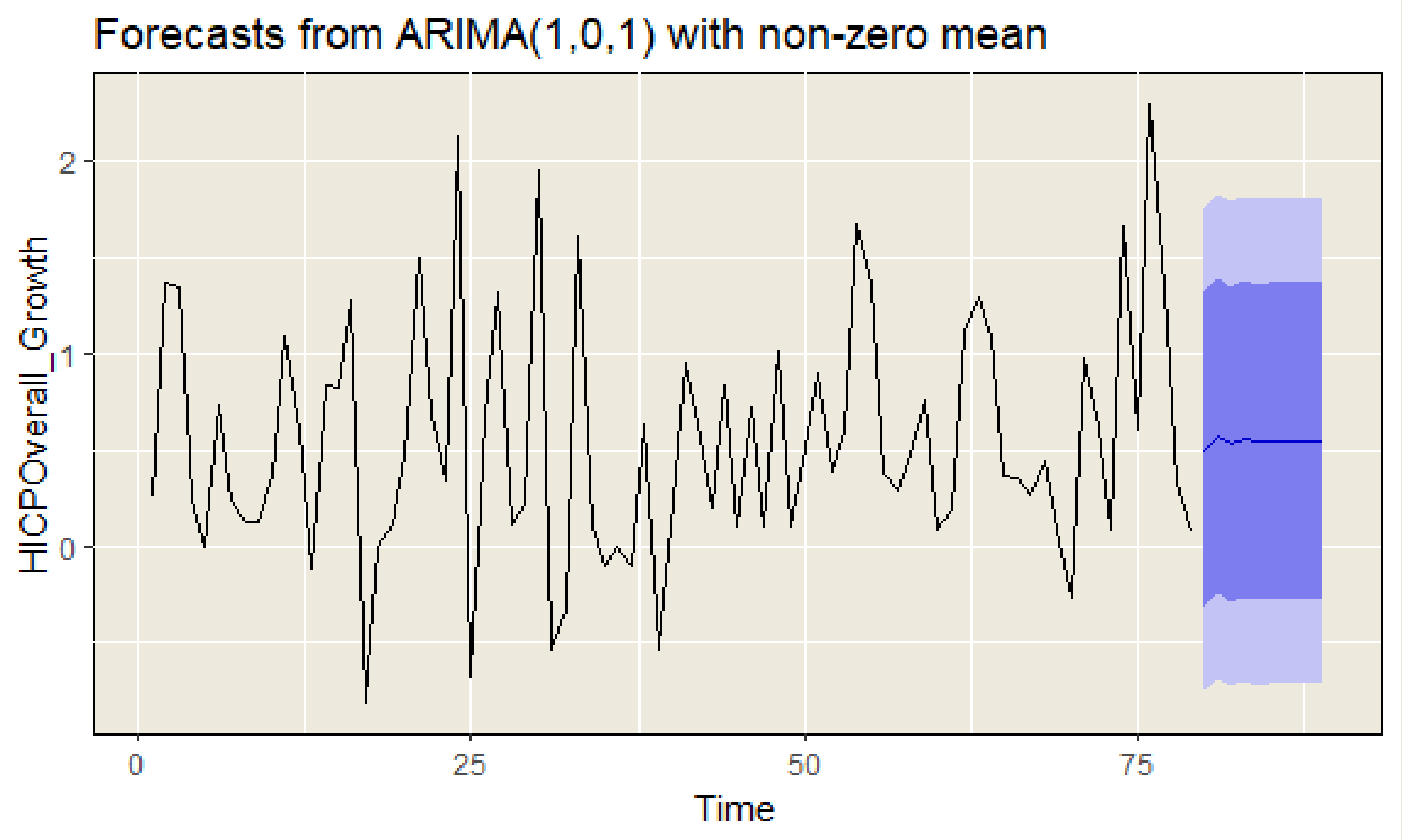
AR1

- **BIC = 163.87**
- **Coeff : ar1:-0.0455 (|*B*| <1)**
- **RMSE : 0.6282927**
- **Point forecast; Yt+1 = 0.5688675**
- **95% CI : (-0.6784537**
- **1.816189)**



ARMA1

- **BIC = 167.79**
- **Coeff : ar1: -0.6177 , ma1: 0.6874**
- **RMSE : 0.62646**
- **Point forecast; Yt+1 = 0.4982422**
- **95% CI : (-0.7535918**
- **1.750076)**

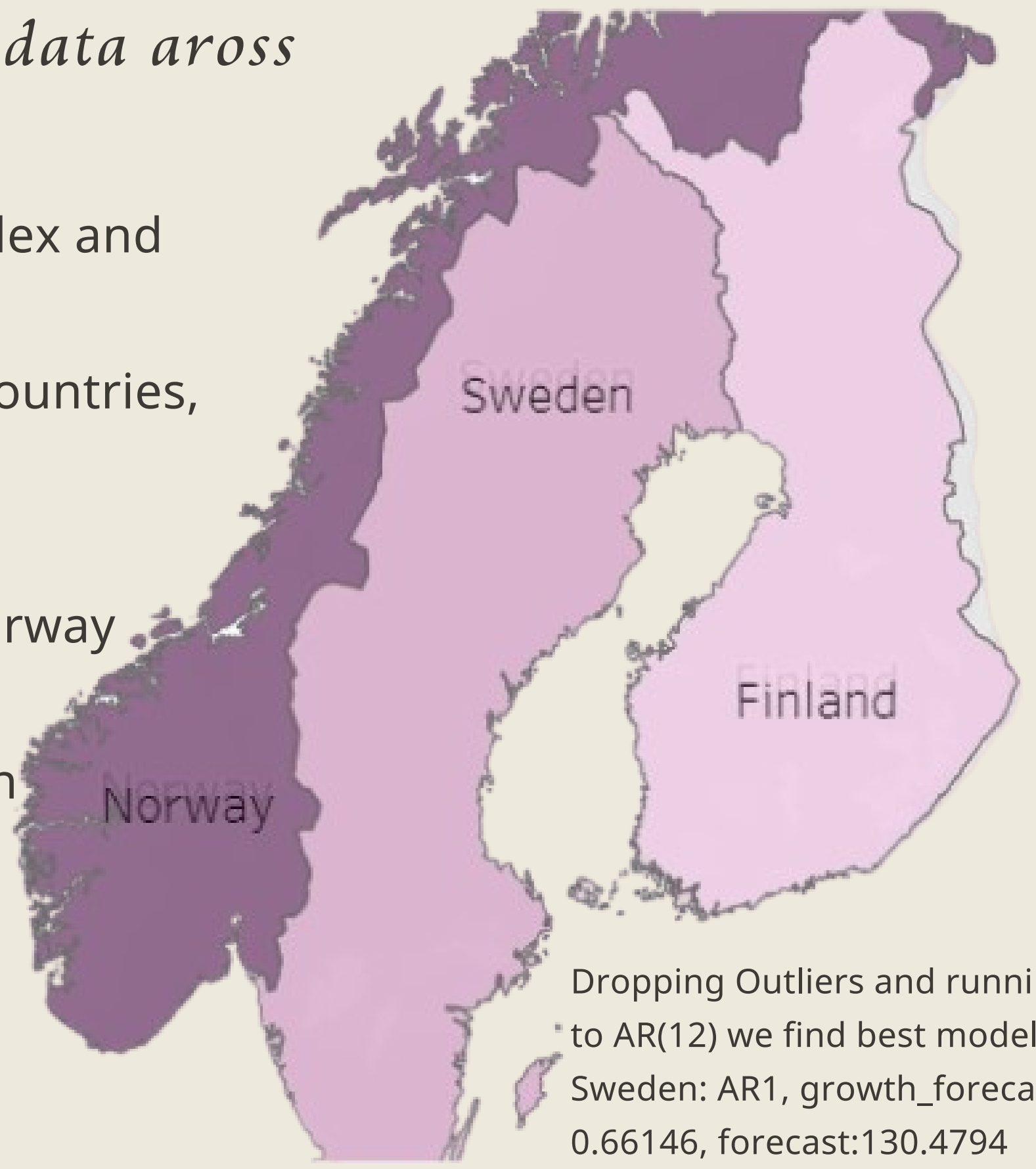


the AR1 model built on data without outliers seems to have the lowest BIC. The AR1 forecasts that the Overall_HICP is *expected to increase by 0.628%* in Yt+1. The expected value of the overall HICP index for Q1 2023 is **136.3756.** At the 95% confidence level this value is **(136.2747, 136.4776)**

03. Comparing HWEG data aross Europe

Let's focus on the HWEG index and create similar forecasts for neighboring Scandinavian Countries, Finland and Sweden**

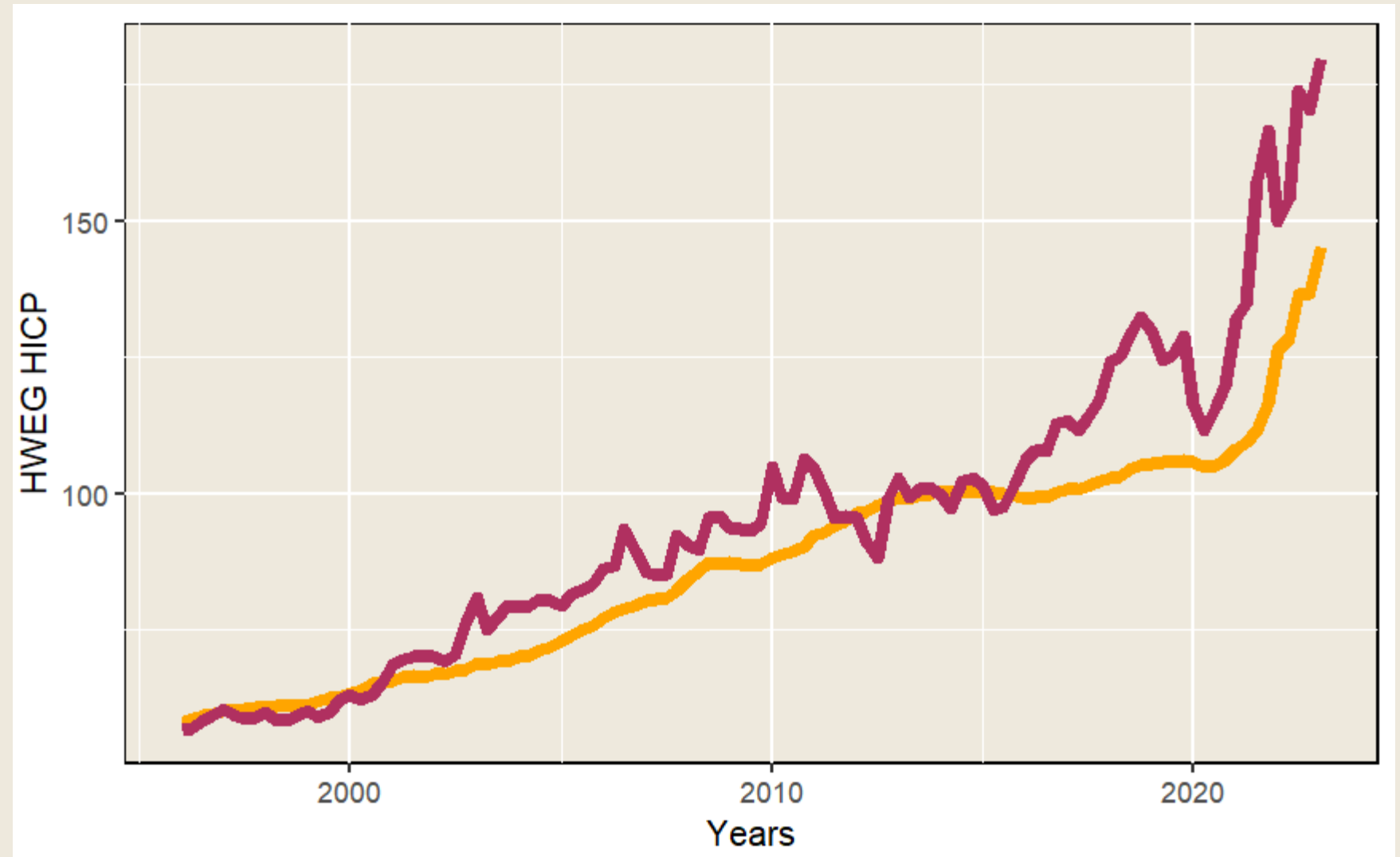
Our model forecasts that Norway can expect energy and housing-driven inflation than its counterparts



Dropping Outliers and running AR(1) to AR(12) we find best models
Sweden: AR1, growth_forecast: 0.66146, forecast:130.4794
Finland: AR1, growth_forecast: 0.236104, forecast:138.2285

3.1 HWEG across the EEA

We also look at the HWEG_HICP index across the rest of the EEA(European Economic Area). The Norwegian HWEG_HICP (maroon) seems to be above EEAs HWEG_HICP(orange). This could be due to higher costs for imported goods on a weaker Norwegian krone. The **0% interest rate** prevalent during the pandemic could also have pushed housing prices higher



We drop outliers from the HWEG HICP for EEA and find AR1 to AR12. AR2 has the lowest BIC so we use the AR2 model to forecast the expected inflation rate for HWEG in the EEA= 0.73459% and the forecasted index value for Q1:2023 = 145.2157

04. Conclusion and Limitations

Our models forecast an expected increase of 0.532% in the HWEG_HICP index and an expected increase of 0.569% in the overall HICP Index. The Norwegian Central Bank abstained from raising interest rates in its January 2023 session. They may consider further tightening to deal with a housing boom, a weakening Kron, and high energy prices.

LIMITATIONS: Our model could have the following limitations:

1. We only look at lags of Yt as predictors. Inflation could be dependent on several external factors. We could improve the forecast by including explanatory variables like interest rates, consumer sentiment, GDP, energy prices, etc. (though we will have to deal with *endogeneity*)
2. The error term may not be a white noise process.

**NOTE: On dropping outliers from Sweden and Finland's dataset we discover the value for Q4 2022 was also an outlier and gets dropped. Hence this map is based on forecasts for Q4 2022, not Q1 2023